Backtesting Report: Momentum vs. Mean Reversion Strategies

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1. Introduction

This report analyzes a backtested trading strategy applied to a multi-instrument dataset. The goal is to compare a **momentum** strategy against a **mean-reversion** strategy, using a buy-and-hold approach as a benchmark. Both strategies were tested on historical price data for multiple instruments over the first eight months of 2024. Key performance metrics are computed to assess returns, risk, and trading efficiency. The analysis highlights strategy behavior and metrics such as cumulative returns, Sharpe ratio, drawdowns, win/loss ratio, and average trade performance.

2. Strategy Overview

The backtest uses data from two provided CSV files: one listing instrument information (project2_instruments.csv) and one containing time-stamped trading prices (project2_trading.csv). These were merged on instrument ID. The dataset spans early January 2024 through mid-August 2024 across multiple instruments. The strategy definitions are:

- **Buy & Hold (Benchmark):** Always fully invested in each instrument's price, with returns simply given by daily price changes.
- **Momentum Strategy:** Uses short-term and long-term moving averages (10-day vs. 50-day). A *buy* (long) signal is generated when the 10-day simple moving average (SMA10) exceeds the 50-day SMA for an instrument, indicating upward momentum.
- **Mean Reversion Strategy:** Defined as the complement of the momentum rule: effectively a buy signal when the price is below the 10-day moving average (oversold condition). In practice this was implemented as taking trades when the momentum signal is *off* (momentum_signal = 0), interpreting those as mean-reversion entries.

The purpose is to see which signal (momentum vs. mean-reversion) yields better risk-adjusted performance, and how they compare to simply holding the instruments.

3. Data Preparation

The dataset preparation involved the following steps:

- **Loading and merging:** The instruments and trading CSV files were read into data frames and merged on instrument_id.
- **Timestamp processing:** All timestamps were parsed as datetime and the merged data sorted by instrument_id and time.
- **Return calculation:** For each instrument, the daily (or per-timestamp) price return was computed via percentage change (with the first entry being NaN). This yielded a new price_return column.
- **Return scaling and clipping:** To avoid extreme effects from very large moves, the raw returns were scaled and clipped. Specifically, returns were divided by 10 (and later by 100) so that no single period's return exceeded ±2%. Also log returns were computed as log(1+price_return) and clipped between -1 and +1 to cap volatility. These steps ensure the exponential cumulation would not "explode" due to outliers. (This has the effect of compressing actual returns into a narrow range for the purpose of the backtest.)
- **Cumulative returns:** Cumulative return series were computed from the adjusted returns: one based on the log-return sum for buy-hold, and later as simple compounded returns (1+return).cumprod() for each strategy after scaling.
- **Cleaning:** Any intermediate NaN values (from the first return or rolling windows) were removed so that each series could be cumulatively multiplied without gaps.

These preprocessing steps yielded a final dataset with columns for the original prices, returns, the two signals, and cumulative return series for each strategy.

4. Indicator and Signal Design

The strategy uses simple moving averages (SMAs) as the technical indicators:

- **SMA10:** the 10-period simple moving average of the instrument's price.
- **SMA50:** the 50-period simple moving average.

The **momentum signal** is defined as:

momentum_signal = 1 if SMA10 > SMA50, else 0.

This means the strategy *goes long* when the short-term trend is above the long-term trend.

The **mean-reversion signal** is defined as the opposite case (price below SMA10):

meanrev_signal = 1 if price < SMA10, else 0.

In execution, this was effectively implemented by using (1 – momentum_signal) to indicate when momentum signal is off. That is, whenever momentum_signal == 0, we treat that as a mean-reversion buying condition (the script sets mean-reversion returns to price return whenever momentum_signal is zero).

Threshold rules: There are no additional thresholds beyond the crossovers (no extra buffer or band). The signals are binary (1 or 0), and any change in sign (from 0 to 1 or vice versa) is interpreted as entering or exiting a position on the next day.

5. Trade Execution Logic

Trades are executed on a daily basis per instrument as follows:

- The momentum and mean-reversion signals are **shifted by one day**. This simulates acting on the signal at the close of day *t* to take a position on day *t+1*.
- On day *t+1*, the strategy's return equals the prior day's signal multiplied by the current day's price return. For example, if momentum_signal(t) = 1, then on day *t+1* the momentum return is price_return(t+1). If the signal is 0, the strategy is out of the market (return = 0). The same logic applies to mean reversion using (1 – momentum_signal).
- **Transaction assumptions:** The backtest ignores slippage and uses discrete daily rebalancing with a one-day delay on signals. A brief fee of 0.05% per trade was calculated but not included in the primary performance summary (so the reported results assume zero transaction cost). The effect of a small fee was examined (computing momentum_ret_cost), but the final metrics are from the cost-free series.
- **Return calculation:** Each strategy's cumulative return series is built by compounding its daily returns:
    - For buy-and-hold, cumulative return = cumulative product of (1 + price_return).
    - For momentum, cumulative return = cumulative product of (1 + momentum_return).
    - For mean reversion, cumulative return = cumulative product of (1 + meanrev_return).
- **Benchmark:** The buy-and-hold return is simply the cumulative product of the raw price returns.

This execution logic effectively assumes investing 100% of capital in the instrument when the signal is on, and 0% when off, without leverage or shorting.

6. Performance Metrics

Key results: The table below summarizes total and annualized returns, Sharpe ratios, maximum drawdowns, win/loss ratios, and average trade performance for each strategy over the backtest period. (Annualized returns assume ~223 calendar days of data, about 0.61 years.)

Strategy	Total Return	Annualized Return (CAGR)	Sharpe Ratio	Max Drawdown	Win/Loss Ratio	Avg Trade Return
Buy & Hold	+0.75%	+1.23%	0.0231	–1.47%	N/A	N/A
Momentum (SMA crossover)	**+1224.96%**	**+6767.36%**	**14.19**	**0.00%**	14061 : 0 (100%)	+2.00%
Mean Reversion	–92.396%	–98.53%	–14.91	–92.39%	0 : 15939 (0%)	–2.00%

- **Cumulative Return:** The momentum strategy's cumulative return far exceeds all others (≈12.25× growth), whereas mean reversion essentially loses 92.4% of capital. Buy & hold barely grows (≈0.75%).
- **Annualized Return:** Momentum's annualized return is ~6767% (reflecting exponential growth in <1 year), buy & hold ~1.23%, and mean-reversion -98.5% (nearly complete loss).
- **Sharpe Ratio:** Momentum has an exceptionally high Sharpe (~14.2) due to near-zero volatility in its compounded series and mostly positive returns. Mean reversion is similarly extreme negative (-14.91). Buy & hold's Sharpe is near zero (0.023).
- **Max Drawdown:** Momentum incurred 0% drawdown (it only trended upward), buy & hold saw a minor –1.47% drawdown, and mean reversion had a –92.39% drawdown (mirroring its total loss).
- **Win/Loss Ratio:** The momentum strategy had 14,061 winning trades and 0 losing trades (≈100% win rate as constructed), whereas mean-reversion had all 15,939 trades losing (0% wins). Buy & hold is not a trading system, so wins/losses are not applicable.
- **Average Trade Return:** The momentum system's average trade return was about +2.00% (reflecting that nearly every trade yielded +2% after scaling), while mean reversion's average was about –2.00%. (These are simple averages of the scaled returns.)

Comparison of Cumulative Returns: The figure below (conceptual) would show the growth of $1 invested in each strategy over time. In tabular form, the final cumulative values (1 +

total return) are ~1.0075 for buy & hold, ~13.2496 for momentum, and ~0.0760 for mean reversion, matching the percentages above.

The momentum strategy dramatically outperformed the others in both return and risk-adjusted metrics. However, as discussed below, these results include strong caveats.

7. Observations and Limitations

Several important caveats and anomalies are evident from the results:

- **Extreme Performance:** The momentum strategy's results are extraordinarily strong (Sharpe ≈14, 1225% gain in 8 months, no drawdown). This suggests an *unrealistic* scenario: essentially, every momentum trade was a winner and there were no losing trades. In practice, such "too good to be true" performance indicates either data quirks or lookahead bias. The mean-reversion strategy's symmetric result (all trades losers, 92% loss) reinforces this.
- **Signal Definition Issue:** The implementation of mean reversion as simply the inverse of the momentum signal may be flawed. In effect, whenever momentum = 0, mean-rev took the full loss of the stock return. This extreme design (and the uniform ±2% scaled returns) creates a situation where one strategy captures all positive moves and the other all negative moves. This perfect segregation inflates Sharpe and win-rate beyond realism.
- **Return Scaling and Clipping:** The returns were arbitrarily scaled (1/10 then 1/100) and clipped to ±2% to "cap extreme swings." While intended to stabilize calculations, this distorts true equity growth. It also normalizes all positive moves to ~2%, so momentum trades always got +2%. In reality, returns vary widely. The scaling choice effectively maximizes Sharpe by compressing volatility.
- **No Trading Frictions:** Transaction costs and slippage were ignored in the main results. Even a small per-trade fee would diminish the momentum strategy's return. The code computed a 0.05% fee but did not apply it to the summary. Realistic execution costs could materially alter these metrics.
- **Lookahead and Rebalancing:** The strategy assumes you can perfectly re-invest in the next day's move by shifting signals by one day. If data is not truly end-of-day, there could be lookahead bias (using tomorrow's information today). The one-day lag mitigates this, but it's unclear if timestamps reflect end-of-day or asynchronous ticks.
- **Data Sampling and Survivorship:** The analysis lumps all instruments' returns together, but it is not clear if the data includes delisted assets or survivorship bias. The presence of huge one-day jumps (±0.02 after scaling) is suspiciously uniform. Real market data would have a wide distribution.

- **Unrealistic Sharpe Values:** A Sharpe ratio of 14 is almost certainly optimistic or erroneous in real markets. It indicates the strategy's distribution of returns was artificially tight and consistently positive. The same for the –14 of mean reversion. In reality, risk (volatility, drawdowns) would be larger.
- **Lack of Risk Control:** The momentum strategy never suffered a drawdown (0% max drawdown), which implies it was effectively risk-free. This is not realistic unless one strategy is magically perfectly aligned with all upward price moves. It suggests overfitting or a definition that captured all positive returns and none of the downside.

In summary, while the raw metrics favor the momentum strategy, they should be viewed cautiously. The backtest assumptions (extreme return clipping, perfect win/loss partitioning, no costs) likely cause **signal instability** and **unrealistic outcomes**. A more conservative approach would reintroduce volatility, use realistic data, and evaluate performance over multiple regimes.

8. Conclusion

The backtest indicates that the **momentum** strategy vastly outperformed the buy-and-hold and the contrarian (mean-reversion) strategy over the sample period. It produced a cumulative gain of ~1225% with an extremely high Sharpe ratio (~14) and no observed drawdown. In contrast, the mean-reversion approach lost ~92% of capital with a Sharpe of –15. The benchmark buy-and-hold returned only about +0.75%.

However, these results hinge on idealized conditions. The momentum strategy's nearly perfect win-rate and zero drawdown suggest an overly optimistic scenario. Key findings include:

- Momentum's returns were driven by capturing every positive scaled move (+2%) and avoiding all negatives, yielding a 100% win rate on trades.
- Mean-reversion ended up taking the opposite side and suffered total losses, highlighting the strategy's poor effectiveness under these definitions.
- The return-scaling and clipping procedure, while preventing numerical issues, created artificially low risk and exaggerated performance metrics.
- Practically, such outcomes are unlikely in a real market setting. The analysis serves more as a proof of concept of the signals rather than a deployable strategy.

Key takeaway: A straightforward SMA-based momentum rule *can* outperform a naive contrarian rule in this backtest, but the magnitude of the outperformance is suspect. The results emphasize the importance of testing backtests for realism: especially by checking drawdown behavior, win/loss distribution, and sensitivity to assumptions. In any

implementation, one should incorporate transaction costs, realistic return variability, and robust validation to avoid being misled by an overly-optimistic simulated strategy.